# Algorithm Analysis - Time complexity :

We have algorithms for searching in memory, and the time growth rate for each is given as follows:

- **Algorithm (A):**

  $f(x) = x^2$ (quadratic growth)

- **Algorithm (B)**

  $g(x) = x\ln(x)$ (linearithmic / log-linear growth)

To compare the growth behavior at infinity, you are required to study the following limits:

$$\lim_{x \to \infty} \frac{f(x)}{g(x)} = \lim_{x \to \infty} \frac{x^2}{x \ln x}$$

Requirements:

1. Simplify the expression inside the limit.
2. Calculate the value of the limit, if it exists
3. Interpret the result in terms of comparing the growth rates of the algorithms, and detemine which on is more computationally expensive for larg value of $x$.

Solution:

1) Simplify the expression:

$$\frac{x^{3}}{x\ln x} = \frac{x \cdot x}{x \ln x} = \boxed{\frac{x}{\ln x}}$$

2) $\frac{x}{\ln x} = \frac{\infty}{\infty} \Rightarrow I.F$

L'Hôpital Rule:

$$\left.\begin{aligned} (x)' &= 1 \\ (\ln x)' &= \frac{1}{x} \end{aligned}\right\} \frac{1}{\frac{1}{x}} = x \Rightarrow x = \infty$$

$$\text{So: } \lim_{x \to \infty} \frac{x^{3}}{x \ln x} = \infty$$

3) Interpretation

- Algorithm A: $O(x^2)$
- Algorithm B. $O(x \ln x)$

Since the ratio tends to infinity, this mean that:
$x^2$ grows much faster than $x \ln x$
that is, Algorithm A is significantly more computationally expensive for large values of $x$, while Algorithm B is more efficient and has slower growth.